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Question Paper Code : 19321

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2022

Third Semester

B.E. – Computer Science and Engineering

19CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

A, CO1 1. A CPU has 32-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)? Justify your answer.

A) 350 B) 325 C) 400 D) 375

U, CO1 2. Write the basic performance equation.

A, CO2 3. Represent the decimal value -95 in 8-bit signed magnitude and 2's complement representation.

A, CO2 4. Represent the number 0.00010101 x 2^{16} using IEEE Single precision format.

U, CO3 5. Which of the following instructions modifies the program counter? Justify your answer.

A) Only Load instruction B) Only ADD instructions

C) Only Branch instructions D) All the given instructions

R, CO3 6. _____ style computers aimed at reducing the time of executing an instruction. Justify your answer.

A) RISC B) ISA C) CISC D) Intel

A, CO4 7. Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the Least Recently Used (LRU) scheme. The number of cache misses for the following sequence of block addresses is 3, 22, 3, 0, 22, 3. Justify your answer.

A) 2 B) 3 C) 4 D) 0

A, CO4 8. How many 64K x 1 RAM chips are needed to provide a memory capacity of 256K-bytes? Justify your answer.

A) 8 B) 32 C) 64 D) 128

U, CO5 9. Shall we connect the I/O Devices directly to the System Bus? Justify your answer.

R, CO5 10. What is DMA? How it is useful?

PART B – (5 × 16 = 80 Marks)

A, CO1 11. a) i. Represent the numbers -106 and 118 using 8-bit Signed Magnitude, 1's complement and 2's complement Representation. Also, perform the arithmetic operation -106 + 118 using 2's complement representation. (8 Marks)

U, CO1 11. a) ii. List and explain different addressing modes used to represent the operands in RISC style computers. (8 Marks)

OR

A, CO1 11. b) i. Write an assembly language program to find the biggest of N numbers stored at the starting memory location LOC. Assume the word size is 32 bits. (8 Marks)

U, CO1 11. b) ii. Write notes on the components involved in the connection between processor and main memory with a neat diagram. (8 Marks)

A, CO2 12. a) i. Write short notes on Booth algorithm and apply Booth's algorithm to multiply two 5-bit signed numbers +13 and -6. (8 Marks)

A, CO2 12. a) ii. Perform multiplication of two 4-bit unsigned numbers 14 and 9 using Sequential circuit multiplication. (8 Marks)

OR

A, CO2 12. b) i. Write short notes on Restoring Division algorithm and apply Restoring division algorithm to perform 8 / 3. (8 Marks)

A, CO2 12. b) ii. Perform multiplication of two 8-bit signed numbers 95 and 22 using Bit recoding of Multipliers. (8 Marks)

U, CO3 13. a) i. Draw and explain the datapath in a processor with 5 stages of execution Instruction Fetch (IF), Instruction Decode (ID), Execution (EX), Memory Access (MEM), Write Result (WR). (8 Marks)

A, CO3 13. a) ii. Write the sequence of actions needed to fetch and execute the following instruction using data path registers:

A) Branch_if_[R5] = [R6] LOOP

B) Load R5, X(R7) (8 Marks)

OR

U, CO3	13. b) i.	Describe in detail about different branch prediction techniques in pipelining to avoid the penalty with necessary diagrams.	(8 Marks)

A, CO3	13. b) ii.	Identify whether data hazard exists in the following code. If so, provide the various ways of overcoming the data hazards.
		ADD R2, R3, #100
		SUB R9, R2, #30
			(8 Marks)

A, CO4	14. a) i.	Describe how to design a large memory structure for a 8 M × 16 memory using 1 M × 4 memory chips with a suitable diagram.	(8 Marks)
U, CO4	14. a) ii.	List out different types of ROMs and highlight their features.	(8 Marks)
		OR	
A, CO4	14. b) i.	A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 eight bit words. For the given cache design, A) How many bits are required for addressing the main memory? B) How many bits are needed to represent the TAG, SET and WORD fields?	(8 Marks)
U, CO4	14. b) ii.	Write a short note on magnetic disks.	(8 Marks)
U, CO5	15. a) i.	Explain the handshake control of data transfer during an input operation using Asynchronous Bus with a neat diagram.	(8 Marks)
U, CO5	15. a) ii.	Describe in detail about Parallel interface circuits for an Input device with its functions.	(8 Marks)
		OR	
U, CO5	15. b) i.	With neat diagrams, explain PCI bus structure and also describe different data transfer signals associated with PCI bus.	(8 Marks)
U, CO5	15. b) ii.	With necessary steps, explain the mechanism for handling and disabling the interrupts.	(8 Marks)